

Dual-input convolutional neural network for glaucoma diagnosis using spectral-domain optical coherence tomography

Sukkyu Sun,¹ Ahnul Ha,^{2,3} Young Kook Kim ,^{2,4} Byeong Wook Yoo,⁵ Hee Chan Kim,^{1,6,7} Ki Ho Park ^{2,4}

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For numbered affiliations see end of article.

Correspondence to

Hee Chan Kim, Department of Biomedical Engineering, Seoul National University Hospital, 101 Daehak-ro, Jongro-gu, Jongno-gu 03080, Korea (the Republic of); and Ki Ho Park, Ophthalmology, Seoul National University College of Medicine, 101 Daehak-ro, Jongro-gu, Seoul 03080, Korea (the Republic of); hckim@snu.ac.kr kihohpark@snu.ac.kr

SS and AH contributed equally to this work as co-first authors.

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ABSTRACT

Background/Aims To evaluate, with spectral-domain optical coherence tomography (SD-OCT), the glaucoma-diagnostic ability of a deep-learning classifier.

Methods A total of 777 Cirrus high-definition SD-OCT image sets of the retinal nerve fibre layer (RNFL) and ganglion cell-inner plexiform layer (GCIPL) of 315 normal subjects, 219 patients with early-stage primary open-angle glaucoma (POAG) and 243 patients with moderate-to-severe-stage POAG were aggregated. The image sets were divided into a training data set (252 normal, 174 early POAG and 195 moderate-to-severe POAG) and a test data set (63 normal, 45 early POAG and 48 moderate-to-severe POAG). The visual geometry group (VGG16)-based dual-input convolutional neural network (DICNN) was adopted for the glaucoma diagnoses. Unlike other networks, the DICNN structure takes two images (both RNFL and GCIPL) as inputs. The glaucoma-diagnostic ability was computed according to both accuracy and area under the receiver operating characteristic curve (AUC).

Results For the test data set, DICNN could distinguish between patients with glaucoma and normal subjects accurately (accuracy=92.793%, AUC=0.957 (95% CI 0.943 to 0.966), sensitivity=0.896 (95% CI 0.896 to 0.917), specificity=0.952 (95% CI 0.921 to 0.952)). For distinguishing between patients with early-stage glaucoma and normal subjects, DICNN's diagnostic ability (accuracy=85.185%, AUC=0.869 (95% CI 0.825 to 0.879), sensitivity=0.921 (95% CI 0.813 to 0.905), specificity=0.756 (95% CI 0.610 to 0.790)) was higher than convolutional neural network algorithms that trained with RNFL or GCIPL separately.

Conclusion The deep-learning algorithm using SD-OCT can distinguish normal subjects not only from established patients with glaucoma but also from patients with early-stage glaucoma. The deep-learning model with DICNN, as trained by both RNFL and GCIPL thickness map data, showed a high diagnostic ability for discriminating patients with early-stage glaucoma from normal subjects.

INTRODUCTION

Glaucoma, a neurodegenerative disease brought on by progressive retinal ganglion cell (RGC) loss, is associated with structural changes of the optic nerve head (ONH) and retinal nerve fibre layer (RNFL).^{1 2} As glaucoma progression runs its course, structural changes often are detected before functional losses

become clearly apparent.³ Therefore, early detection of glaucomatous structural damage is essential for proper management and maintenance of patients' vision-related quality of life.⁴

Spectral-domain optical coherence tomography (SD-OCT) is commonly employed as an imaging modality for evaluation of glaucomatous structural damage.⁵ Its clinical utility for glaucoma is primarily in the evaluation of RNFL parameters for provision of comprehensive assessments of RGC axons approaching the ONH.^{6 7} Furthermore, SD-OCT's enhanced performance allows for assessment of macular parameters in glaucoma evaluation, which utility is especially valuable, as the macula contains the highest RGC concentration in the retina⁸; thus too, RGC loss typically is more readily detectable in the macula.⁹

The most effective and widely used deep-learning algorithm for images and videos is convolutional neural network (CNN). CNN allows for both extraction of features from image data in the convolutional layers and pooling layers, and classification based on the fully connected layers' features.¹⁰ If multi-domain feature extraction can be combined with deep learning, sufficient information can be provided for the classification task, and better results are likely to be achieved. Recently, the dual-input convolutional neural network (DICNN) has shown great utility as well as potential as a deep-learning algorithm for effective classification based on two images used as inputs. Li *et al*¹¹ showed that a dual-input network structure using both ECG and phonocardiogram signals identified more underlying features inside the signals, thereby improving the performance for diagnosing coronary artery disease. Likewise, Choi *et al*¹² developed a dual-input deep-learning model that interprets two different radiographs simultaneously for detection of supracondylar fracture, showing a diagnostic accuracy comparable to radiologists'. This approach is promising, as it mimics how a human clinician interprets images in making a diagnosis.

In this paper, we propose this deep-learning algorithm as an effective and efficient means of diagnosing early glaucoma accurately based on two images. DICNN was trained with both RNFL and macular ganglion cell-inner plexiform layer (GCIPL) thickness maps that enabled it to discriminate between normal and glaucomatous eyes. DICNN was evaluated along with other methods for accuracy and area under the receiver operating characteristic (ROC) curve (AUC).

METHODS

Study subjects

Eyes were chosen from a database of subjects examined for glaucoma at the Glaucoma Clinic, Seoul National University Hospital, between January 2012 and January 2019. This study was approved by the Seoul National University Hospital Institutional Review Board and adhered to the tenets of the Declaration of Helsinki. Informed consent was waived due to the study's retrospective nature.

All of the study subjects underwent a complete ophthalmic examination that included best-corrected visual acuity, refraction, slit-lamp biomicroscopy, gonioscopy, Goldmann applanation tonometry (Haag-Streit, Koniz, Switzerland), dilated stereoscopic examination of optic disc, digital colour stereo disc photography (SDP), red-free RNFL photography, Cirrus HD-OCT and central 24-2 threshold testing of the Humphrey visual field (HVF) (HFA II; Humphrey Instruments, Dublin, CA, USA). The patients additionally underwent, at the baseline examination, central corneal thickness measurement (Orbscan 73 II, Bausch & Lomb Surgical, Rochester, NY, USA) and axial-length measurement (IOL Master ver. 5, Carl-Zeiss Meditec, Dublin, CA, USA).

For inclusion in the study, patients with glaucoma had to satisfy the definition of primary open-angle glaucoma (POAG), irrespective of untreated intraocular pressure (IOP) level: the presence of certain characteristic changes (ie, localised or diffuse neuroretinal rim thinning/notching) on SDP; RNFL defects in corresponding regions on red-free RNFL photographs with open anterior chamber angles. The POAG diagnosis was made regardless of presence or absence of glaucomatous visual field (VF) defect. See online supplemental text for definition of glaucomatous VF defect and detailed exclusion criteria of the study subjects.

Final inclusion or exclusion of patients was determined by an experienced ophthalmologist (YKK). Since our outcomes were measured at the ocular (eye) level, we tried to minimise problems that could arise from violating the assumption made by the majority of statistical tests that each data value is independent. In cases where both eyes proved eligible, therefore, one eye was selected randomly. Subsequently, the patients with POAG were divided into two groups, based on the standard automated perimetry (HVF analyzer 24-2 SITA-Standard strategy) data: early glaucoma (VF mean deviation (MD) > -6 dB) and moderate-to-severe glaucoma (VF MD < -6 dB).

The normal eyes in this study were from subjects without any ocular abnormalities in either eye, since patients with unilateral glaucoma constitute a group at higher risk for future development of glaucoma in the other, normal eye.¹³ They met the following criteria: (1) baseline IOP less than or equal to 21 mmHg, with no IOP-elevation history; (2) no glaucomatous optic disc changes as detected on SDP; (3) no RNFL defects on red-free fundus images; and (4) normal HVF results.

OCT imaging

All scans were acquired according to the Optic Disc Cube 200 $\times$ 200 protocol of Cirrus HD-OCT, which was designed for cube-scan positioning on the ONH, and which is the primarily used modality for glaucoma analysis. Once the subject was seated and properly aligned, the iris was brought into view by means of a mouse-driven alignment system, and the line-scanning ophthalmoscopic image was brought into focus with adjustment for refractive error. Then, the ONH was centred in the live image, and the centring (Z-offset) and enhancement (polarisation) were optimised. The laser subsequently scanned a 6 mm $\times$ 6 mm area, thus capturing a data cube consisting of 200 A-scans from 200 linear B-scans (40 000 points), all in approximately 1.5 s (at

27 000 A-scans/s). The ONH parameters were measured automatically using a Carl Zeiss Meditec analysis algorithm that had been developed for Cirrus HD-OCT (version 6.0). For the macular cube, a 512 $\times$ 128 grid that consists of 128 horizontal B-scans each comprising 512 A-scans was used. Only high-quality scans (signal strength ≥ 7 , with absence of discontinuity or misalignment, involuntary saccade, blinking artefacts, segmentation failure or artefacts) were used in the final analysis.

Dual-Input Convolutional Neural Network

Deep-learning-based features were extracted by training CNN on either peripapillary RNFL or macular GCIPL thickness map images. The images were extracted from the average RNFL thickness and RNFL quadrants, and then were used for analysis of glaucoma on the RNFL thickness map.^{14–16} For medical images with only a small number of data sets, CNN analysis proceeds in two ways (learns RNFL and GCIPL separately), given both the computational expensiveness of the deep-learning algorithm and the likelihood of overfitting of extracted features. However, DICNN can prevent overfitting while using the features of both RNFL and GCIPL simultaneously.

The RNFL thickness map provides a colour-coded display of RNFL thickness within the range of 0–370 μ m. In the present study, the value of each thickness map pixel was normalized to the RNFL thickness, and that value was converted to a value within the 0–255 range for better image representation. Images were resized to 224 $\times$ 224 $\times$ 3 for learning by DICNN. Since a large number of images must be available for better training of CNN, augmentation, which entailed enlargement of image data by addition of Gaussian noise, was performed. Learning was set at 0.01 (gradient descent optimizer).¹⁷ Since the last-fullyconnected layer must have three classes (ie, normal, early POAG, moderate-to-severe POAG), other fully connected layers had 120, 64, 16 and each 16 features from last-fullyconnected layer (RNFL and GCIPL) was reduced to three nodes (three classes).

DICNN proceeded by extracting features selected for deep learning (figure 1). The fully connected layer combines all of the features that were extracted from the convolutional and pooling layers, and analyses the relationships among features in

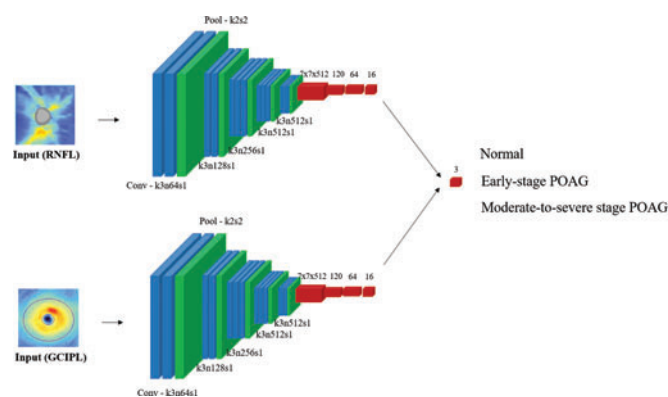


Figure 1 Dual-input convolutional neural network's architecture. The kernel size (k), number of feature maps (n) and stride (s) correspondingly are indicated for each of the convolutional layers. The extracted features from the convolutional layers in each image (ie, RNFL and GCIPL thickness maps) were flattened and fed to successive fully connected layers consisting of 120, 64, 16 nodes. The features from each convolutional layer were fused at the final layers. Three-layer multi-layer perception was adopted as the final classification task for reduction to three nodes. GCIPL, ganglion cell-inner plexiform layer; POAG, primary open-angle glaucoma; RNFL, retinal nerve fibre layer.

the neural network architecture. DICNN was trained by back-propagating the output logit value simultaneously with the training of the weights of the visual geometry group (VGG16). The extracted features from the convolutional layers in each image (ie, RNFL and GCIPL thickness maps) were flattened and fed to successive fully connected layers consisting of 120, 64, 16 nodes.¹⁸ The features from each convolutional layer were fused at the final layers. Three-layer multi-layer perception was adopted as the final classification task for reduction to three nodes.

Hardware specifications

CPU: Intel core i7-7700 3.60 Hz
GPU: TITAN X (Pascal) 12 GB
RAM: 16 GB

Software specifications

Deep-learning libraries
Tensorflow—1.14.0 with cuda 10.0 and cudnn 7.6.3
Python—version 3.6

Statistical analysis

The demographic data were compared among the three groups with the Bonferroni post hoc test by one-way analysis of variance. The categorical data were analysed by χ^2 test with Bonferroni correction. We used the AUC with 95% CIs in the evaluations of the algorithms' performances. The 95% CIs were calculated by applying, out of each first and second fully connected layer of the VGG19, 20% of drop.¹⁹ The sensitivity and specificity in the ROC curves were obtained by thresholding the logit value coming out of the network output.²⁰ For comparison between the two groups, the logit values were normalised to make the sum of their respective logit values to 1. All of the statistical analyses were performed with Matlab (The MathWorks, Natick, MA, USA).

RESULTS

An overview of data sets is shown in online supplemental efigure 1. The data set used in this study (training and testing) included 777 eyes of 777 subjects (462 eyes of 462 subjects with POAG and 315 eyes of 315 normal subjects). Among the 462 POAG eyes, 219 were early-stage, and 243 were moderate-to-severe stage. For the training data set, 621 image sets in total (252 normal, 174 early POAG, 195 moderate-to-severe POAG) were used. During the testing, 156 independent image sets (63 normal, 45 early-stage glaucoma, 48 moderate-to-severe-stage POAG) were assessed by each of the deep-learning algorithms. The subjects' demographic and ocular characteristics are provided in online supplemental efigure S1.

The accuracy and AUC calculated for DICNN are shown in table 1, which also indicates the results obtained by the trained VGG16 with RNFL or GCIPL, respectively. In addition, the accuracy and AUC of the fused-output RNFL VGG16/GCIPL VGG16 ensemble model were shown. The ROC curves for all of the analysis algorithms are plotted in figure 2. The accuracy and AUC were 92.793% and 0.957 (95% CI 0.943 to 0.966), the sensitivity was 0.896 (95% CI 0.896 to 0.917) and the specificity was 0.952 (95% CI 0.921 to 0.952) for DICNN's distinguishing between normal and glaucoma subjects. As for VGG16 with RNFL, the accuracy and AUC were 94.595% and 0.978 (95% CI 0.954 to 0.978), the sensitivity was 0.938 (95% CI 0.771 to 0.938) and the specificity was 0.952 (95% CI 0.937 to 0.981); for VGG16 with GCIPL, the accuracy and AUC were 88.288% and 0.921 (95% CI 0.861 to 0.921), the sensitivity was 0.833 (95% CI 0.614 to 0.833) and the specificity was 0.921 (95% CI 0.857

Table 1 Accuracy and the area under the receiver operating characteristic curve of machine learning algorithms distinguishing between normal subjects and subjects with glaucoma

Normal versus POAG Accuracy (%)	Normal versus early POAG	
	AUC	Accuracy (%)
DICNN (RNFL/GCIPL)	92.79	0.957
VGG16 (RNFL)	94.59	0.978
VGG16 (GCIPL)	88.29	0.921
Ensemble model	92.80	0.966

AUC, area under the receiver operating characteristic curve; DICNN, dual-input convolutional neural network; GCIPL, ganglion cell-inner plexiform layer; POAG, primary open-angle glaucoma; RNFL, retinal nerve fibre layer; VGG, visual geometry group.

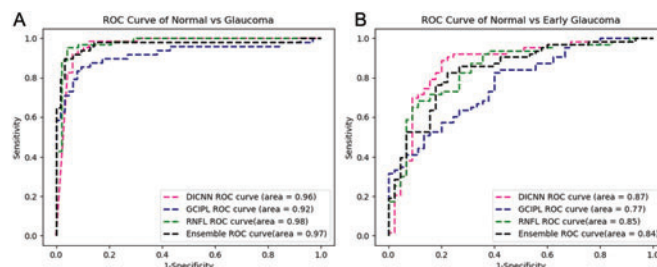


Figure 2 Receiver operating characteristic (ROC) curve of DICNN with both retinal nerve fibre layer (RNFL) and ganglion cell-inner plexiform layer (GCIPL), CNN with single image (RNFL or GCIPL), and the ensemble model (fused output ensemble of VGG16 RNFL and VGG16 GCIPL): (A) distinguishing between normal subjects and patients with glaucoma; (B) between normal subjects and patients with early-stage glaucoma. DICNN, dual-input convolutional neural network; VGG, visual geometry group.

to 0.933). As for the ensemble model, the accuracy and AUC were 92.800% and 0.966 (95% CI 0.943 to 0.972), the sensitivity was 0.875 (95% CI 0.821 to 0.917) and the specificity was 0.968 (95% CI 0.856 to 0.984).

For DICNN's distinguishing between normal and subjects with early-stage glaucoma, the accuracy and AUC were 85.185% and 0.869 (95% CI 0.825 to 0.879), the sensitivity was 0.921 (95% CI 0.813 to 0.905) and the specificity was 0.756 (95% CI 0.610 to 0.790); for VGG16 with RNFL, the accuracy and AUC were 80.0% and 0.850 (95% CI 0.739 to 0.850), the sensitivity was 0.905 (95% CI 0.877 to 0.952) and the specificity was 0.644 (95% CI 0.356 to 0.644); for VGG16 with GCIPL, the accuracy and AUC were 72.900% and 0.767 (95% CI 0.721 to 0.784), the sensitivity was 0.839 (95% CI 0.742 to 0.790) and the specificity was 0.578 (95% CI 0.556 to 0.639). As for the ensemble model, the accuracy and AUC were 72.222% and 0.840 (95% CI 0.648 to 0.856), the sensitivity was 0.921 (95% CI 0.787 to 0.952) and the specificity was 0.444 (95% CI 0.278 to 0.514). The validation accuracies obtained by other optimisers (ie, Adam,²¹ Adadelta²² and Adagrad²³) are shown in online supplemental efigure 2.

Figure 3 shows representative cases of DICNN's misclassification between normal subjects and patients with early-stage POAG. The first row in the figure shows the findings for early-stage POAG that were misdiagnosed as normal images (figure 3A–E). As can be seen, inferotemporal RNFL defect had a relatively long angular distance from the macula. The second row shows the results for a normal subject that were misclassified as early-stage POAG (figures 3F–J). The course of the major temporal blood vessels was more temporal, and thus, the RNFL thickness plot peaks were

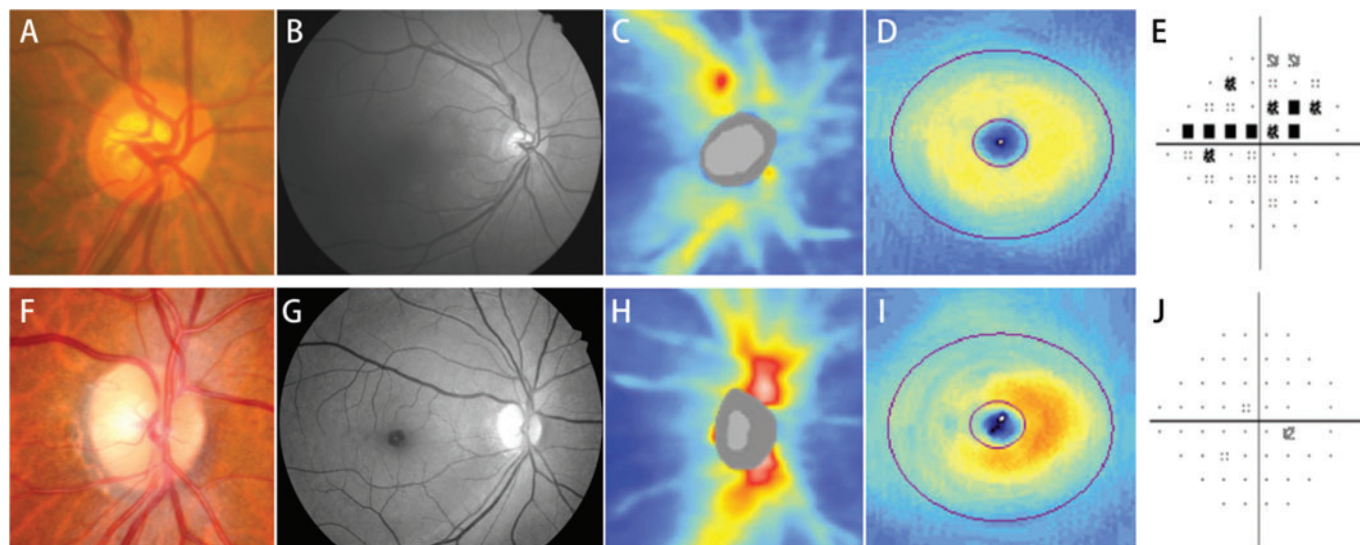


Figure 3 Representative cases of DICNN's misclassification between patients with early-stage POAG (A–E) and normal subjects (F–J). The first row shows the early-stage POAG findings misdiagnosed as normal images by DICNN. Inferotemporal optic disc rim narrowing (A) along with RNFL defect (B) with corresponding superior visual field defect (E) are shown. On that basis, a diagnosis of glaucoma was made. Note that in fact, the inferotemporal RNFL defect had a relatively long angular distance from the macula. The second row shows the normal subject misclassified as early-stage POAG by DICNN. The major temporal blood vessels were more temporally located, thereby causing shifting of peaks on the RNFL thickness plot. DICNN, dual-input convolutional neural network; POAG, primary open-angle glaucoma; RNFL, retinal nerve fibre layer.

correspondingly shifted. This might have caused thinning, as indicated on RNFL and GC IPL thickness maps, relative to the built-in normative database.

DISCUSSION

In the present study, we determined that DICNN combined with features of RNFL and GC IPL thickness was effective in distinguishing glaucomatous from normal SD-OCT images. DICNN was developed by combining the features of both RNFL and GC IPL extracted from last-fully connected layer from VGG16. Notably, DICNN differentiated between subjects with early-stage glaucoma and normal subjects more accurately than did the other methods.

VGG16 with RNFL or GC IPL, respectively, showed similar results for differentiation of patients with glaucoma and normal subjects to a certain extent, though the data amount and glaucoma-diagnostic criteria differed from the previous research.^{14–16 24} Muhammad *et al*²⁵ used a hybrid deep-learning method according to a single wide-field OCT protocol. Their combination of RNFL and GC IPL probability images, however, did not yield diagnostic performance that was superior to that of RNFL probability alone. They suggested that a novel method of combining RNFL with GC IPL information could be a feasible means of enhancing diagnostic performance. In our study, the deep-learning algorithms analysed by DICNN with combined RNFL and GC IPL features on thickness maps showed better diagnostic performance for early-stage glaucoma relative to the conventional deep-learning methods.

Early detection of glaucoma is essential, certainly, but early-onset glaucoma is much more challenging to diagnose than is advanced glaucoma, since early structural damage sometimes is revealed in one parameter but not another. Wang *et al*²⁶ reported that their RNFL thickness evaluation missed 47 (77%) of the 61 eyes showing abnormal thinning on macular scans. The typical analyses of peripapillary RNFL thickness alone can overlook glaucomatous macular damage.^{9 27} On the other hand, it is known that macular GC IPL parameters are less sensitive in

cases where the temporal margin of the defect is located far from the fovea.²⁸ Therefore, in terms of detecting subtle early-glaucomatous damage, more combined information is needed. In fact, based on our results, the features extracted by both RNFL and GC IPL boosted the classification results for early-stage glaucomatous eyes by an average MD -2.83 dB.

The higher accuracy along with the larger AUC of the network trained by DICNN compared with VGG16 (figure 2) was perhaps owed to the DICNN's having more diverse features for analysis of patients with early-stage glaucoma and normal subjects. The RNFL and GC IPL thickness maps present 'related but different' clinical information. DICNN integrated variables from the RNFL and GC IPL thickness maps for training in order to identify more underlying features inside the images. Since the final output values of the fused nodes were trained through backpropagation with softmax cross-entropy loss, the entire network could share variables in both the RNFL and GC IPL thickness maps. The fused features in the last 16 fully connected layers (fully connected layer of 120, 64, 16) of DICNN had the best accuracy and AUC, as shown in figure 2. If there were too many last-fully connected layers (above 16), DICNN would not converge. On the other hand, with too few fully connected layers, the trained network would show poor classification accuracy. With DICNN, the network converges more reliably than with VGG16 trained on either RNFL or GC IPL (figure 4).

Our study has some advantages relative to the previous work. First, we used a class-balanced data set for the early-stage glaucoma, moderate-to-severe stage glaucoma and normal groups. In that way, we were able to minimise the problems incurred from biased training and estimation.²⁵ Second, we used both RNFL and GC IPL information for training of the deep-learning model. As structural damage sometimes is revealed in only one parameter, not also in the other, it is important that in glaucoma diagnostics, information is gathered for both the RNFL and GC IPL.²⁹ Likewise, a deep-learning model that is trained with both RNFL and GC IPL images would be expected to offer better diagnostic performance.

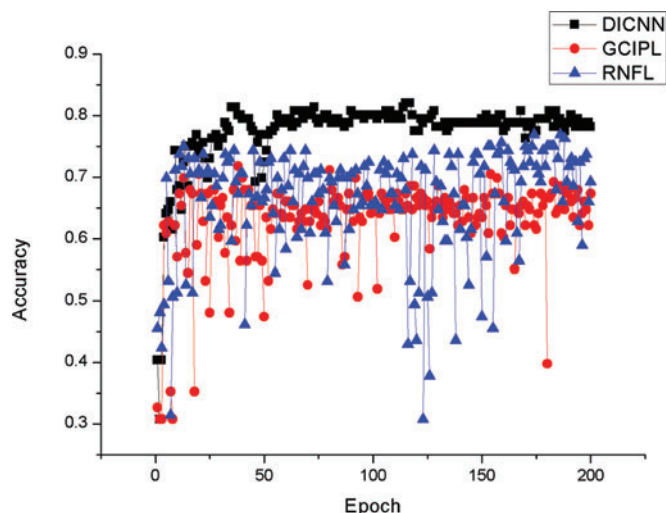


Figure 4 Training curves for dual-input convolutional neural network (black square) and convolutional neural network with ganglion cell-inner plexiform layer (red dot) or retinal nerve fibre layer (blue triangle), respectively.

Our study has several limitations. First, the training results were based on a relatively small amount of data. In the current study, we increased the size of the data set by augmentation (ie, by adding Gaussian noise), thereby improving the network's diagnostic accuracy for glaucoma.³⁰ However, geometric transformation techniques such as flip, rotation or inversion were not applied. Clinically, the specific location of optic neuropathy is an important basis for glaucoma diagnosis, since glaucomatous optic neuropathy generally begins in the inferotemporal disc region and then progresses to the superotemporal and other regions.³¹ We therefore applied only the Gaussian noise, which preserves the spatial dimensions of the images.³² Further studies will validate this algorithm using a larger data set. Second, the performance of our deep-learning algorithm is dependent on the quality of the training images. Exclusion of low-quality images from the training set could limit the usefulness of the algorithm in actual clinical settings. Third, we used VGG16 as the baseline model, since it showed higher accuracy than VGG19's (online supplemental efigure 3). It has been reported that deeper networks usually have better classification performance. The reasons for the different results in our study are not clear, but the number of images is one of the possible ones, since the improved performance of deeper models might not be evident in smaller data sets.³³ The performance differences of different neural network architectures in larger data sets should be evaluated in further studies. Fourth, it remained unclear how the deep-learning model discriminated cases as either glaucoma or normal. There have been many attempts to visualise exactly what a deep-learning network learns, such as by creating heat maps,^{34 35} which process will certainly be applied in a future clinical study.

In conclusion, we constructed a deep-learning model incorporating DICNN as trained by both RNFL and GCIPL thickness map data, and compared and validated the high diagnostic ability for discrimination of patients with early-stage glaucoma from normal subjects. It might be possible to further improve the model's performance by increasing the size as well as diversification of the training set.

Author affiliations

¹Interdisciplinary Program in Bioengineering, Graduate school, Seoul National University, Seoul, South Korea
²Department of Ophthalmology, Seoul National University College of Medicine, Seoul, South Korea
³Department of Ophthalmology, Jeju National University Hospital, Jeju-si, South Korea
⁴Department of Ophthalmology, Seoul National University Hospital, Seoul, South Korea
⁵Samsung Research, Samsung Electronics, Seoul, South Korea
⁶Institute of Medical and Biological Engineering, Medical Research Center, Seoul National University, Seoul, South Korea
⁷Department of Biomedical Engineering, Seoul National University College of Medicine, Seoul, South Korea

Contributors Study design: SS, AH, YKK, HCK, KHP. Writing the article: SS, AH. Data collection: AH, YKK, BWY, KHP. Analysis and interpretation of the data: SS, YKK, BWY, HCK. Literature search: SS, AH, BWY. Critical revision of the article: SS, AH, YKK, HCK, KHP. Final approval of the article: HCK, KH.

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ORCID iDs

Young Kook Kim <http://orcid.org/0000-0002-6037-8449>

Ki Ho Park <http://orcid.org/0000-0002-8137-8843>

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